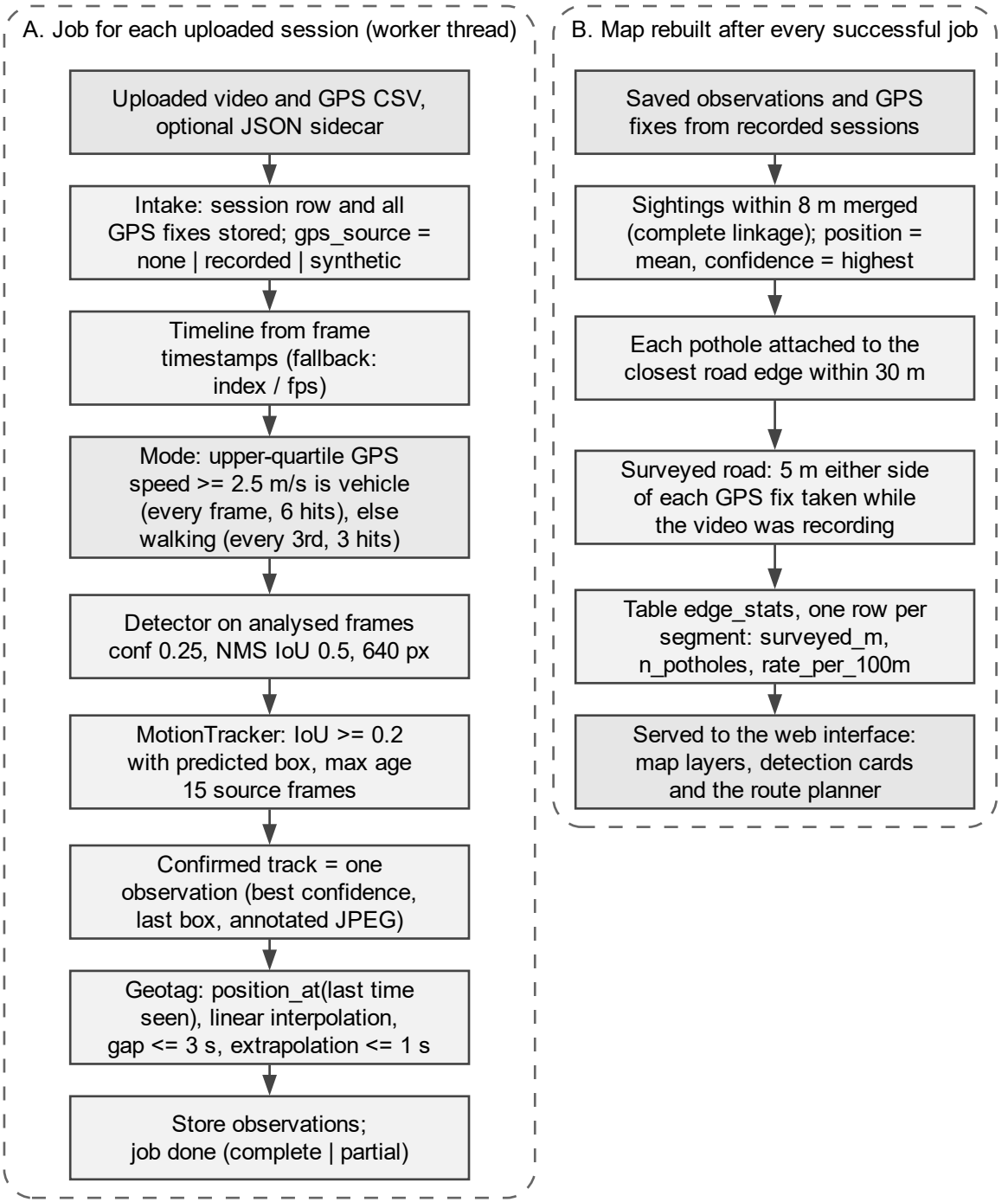
A. Job for each uploaded session (worker thread)
Uploaded video and GPS CSV, optional JSON sidecar
Intake: session row and all GPS fixes stored; gps_source = none | recorded | synthetic
Timeline from frame timestamps (fallback: index / fps)
Mode: upper-quartile GPS speed >= 2.5 m/s is vehicle (every frame, 6 hits), else walking (every 3rd, 3 hits)
Detector on analysed frames conf 0.25, NMS IoU 0.5, 640 px
MotionTracker: IoU >= 0.2 with predicted box, max age 15 source frames
Confirmed track = one observation (best confidence, last box, annotated JPEG)
Geotag: position_at(last time seen), linear interpolation, gap <= 3 s, extrapolation <= 1 s
Store observations; job done (complete | partial)
B. Map rebuilt after every successful job
Saved observations and GPS fixes from recorded sessions
Sightings within 8 m merged (complete linkage); position = mean, confidence = highest
Each pothole attached to the closest road edge within 30 m
Surveyed road: 5 m either side of each GPS fix taken while the video was recording
Table edge_stats, one row per segment: surveyed_m, n_potholes, rate_per_100m
Served to the web interface: map layers, detection cards and the route planner